

Skills Summary

Writing Equations from Word Situations

Use this reference while completing your worksheet or when studying.

Start by naming the letter. Write “let n be ...” with the unit included. Every later step depends on knowing exactly what the letter counts.

Then sort the numbers into two jobs. A *rate* happens for each one of something and is multiplied by the letter. A *one-time amount* (a fee, a starting value, a coupon) is added or subtracted once.

Read your equation back as a sentence. If it does not say what the story says, the model is wrong no matter how neatly it solves.

1. One-step equations from a story

A one-step story does one thing to the unknown: equal groups multiply ($rn = \text{total}$), and adding or removing an amount shifts it ($x - a = b$).

Solve by undoing that single operation on both sides, then check the value in the original sentence.

Example: Boxes of crayons each hold 8 crayons. A shipment holds 96 crayons in all. How many boxes?

Step 1. Equal boxes means one repeated amount, so the number of boxes multiplies the size of a box — a single multiplication step.

Step 2. Let n be the number of boxes: $8n = 96$, so $n = 96 \div 8$.

The shipment holds **12** boxes.

Try it: Trays each hold 7 muffins, and a bakery made 91 muffins. How many trays did it fill?

check: 13

2. Two-step equations from a story

A two-step story has both jobs at once: a one-time amount and a rate. The pattern is *one-time amount* + *rate* $\times$ *letter* = *total*.

Solve in reverse story order: undo the one-time amount first, then undo the rate. A division story ($\frac{x-a}{b} = c$) is the same idea with the two operations swapped.

Example: A bike rental costs 8 dollars to start plus 4 dollars an hour. Dana paid 28 dollars. For how many hours?

Step 1. The starting charge is paid once and the hourly charge repeats, so the rate is the number attached to the letter.

Step 2. Let h be the hours: $8 + 4h = 28$, so $4h = 20$ and $h = 20 \div 4$.

Dana rented the bike for **5** hours.

Try it: A kayak costs 6 dollars to launch plus 5 dollars an hour, and Sam paid 41 dollars. For how many hours?

check: 7

3. Writing an expression and evaluating it

When a story asks for a value rather than an unknown, build the same one-time-amount-plus-rate rule and then substitute the number given.

Growing situations add the rate; shrinking ones subtract it. Substitute into the rule, not into a half-finished computation.

Example: A printer has already produced 25 pages and prints 15 more each minute. How many pages after 6 minutes?

Step 1. The finished pages are there from the start and the printing rate repeats each minute, so the rate multiplies the time and the start is added on.

Step 2. Pages after t minutes: $25 + 15t$. At $t = 6$ this is $25 + 15(6) = 25 + 90$.

There are **115** pages.

Try it: A tank holds 40 litres and loses 6 litres each hour. How much is left after 5 hours?

check: 10

(!) Watch out: the most common modelling error is attaching the letter to the one-time amount instead of the rate. Test your equation with an easy number: if the letter is zero, the equation should give the starting amount.